MIL-R-50792A (AR)
30 OCTOBER 1979
SUPERSEDING
MIL-R-50792 (MU)
31 January 1973

MILITARY SPECIFICATION

ROTATING BAND TUBING, VULCANIZED FIBER

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope.-This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of tubes made of vulcanized fiber for use as rotating bands on 105MM Kinetic energy tank ammunition.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents.-The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-L-10547 - Liners, Case and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-B-46506 - Boxes, Ammunition, Packing Wood Wirebound

FEDERAL

PPP-B-585 - Box, Wood, Wirebound
PPP-B-587 - Boxes, Wood, Wirebound Pallet Type
PPP-B-591 - Boxes, Shipping, Fiberboard, Wood-Cleated

FSC: 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunitions Components

DRAWINGS

11738348 (SH.1) - Band, Rotating (Blank)
11738348 (SH.2) - Band, Rotating (Machined)
11738369 - Band, Rotating

3. REQUIREMENTS

3.1 Material.-The material shall be made by combining layers of chemically gelled paper. The chemical compound used in gelling the paper is subsequently removed by leaching, and the resulting product after being dried and finished by calendering, shall be a dense material of partially regenerated cellulose in which the fibrous structure is retained.

3.2 Zinc chloride.-The fiber tubes shall contain no more than 0.1 percent of zinc chloride.

3.3 Ash content.-The percentage of ash content or mineral matter in the fiber tubes shall not be more than 1.5 percent.

3.4 Color.-The vulcanized fiber tubes shall be furnished in their natural color which is the result of the normal production process. The use of dyes and pigments is permitted with prior approval by the government. Dyes and pigments that mask defects will not be permitted.

3.5 Moisture content.-Moisture content of the fiber tubes shall be 6%, maximum.

3.6 Density.-The fiber tubes shall have a density of 1.38 gms/cc, minimum.

3.7 Hardness.-The fiber tubes shall have a Rockwell Hardness (R scale) of R90, minimum.

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3.8 Compressibility.-Compressibility of a flat .50 in. X .50 in. X .25 in. (12.7mm X 12.7mm X 6.35mm) thick specimen shall be 9% ± 3% of thickness, under a load of 4400 pounds (1996 Kg.).

3.9 Shear strength.-Shear strength of the fiber tubes shall be 4000 psi (27.6 MPa), minimum, in a direction parallel to, or following the curvature of laminations.

3.10 Machining ability.-The quality of the fiber tubes shall be such that it can be readily machined.

3.11 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.12 Workmanship.-The fiber tubes shall be clean and free from contamination (dirt, grease, foreign matter, etc). Also, they shall be free from discoloration, twisting, splitting, cracking, delamination and abnormal warping. The fiber shall be uniform in physical and mechanical properties. See 6.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of six tubes.

4.3.2 Inspection to be performed.-See MIL-A-48078 and paragraph 4.4.2.1 herein.

4.3.3 Rejection.-See MIL-A-48078.

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition one lot of vulcanized fiber tubes is defined as the total quantity of tubes that has been leached in a consistent fashion and subsequently dried as one segregated unit of material (see 6.5). Lot size shall not exceed 600 tubes. Each tube is to be marked with a leaching batch identification code relating to date of manufacture using .25 in (6.35mm) nominal characters on the outer surface at each end within 8 in. (203.2mm) from the ends. The method of marking used shall not result in damage of the finish machined rotating band.

4.4.2 Examination.-See MIL-A-48078.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-R-50792A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Rotating Band Tubing	1 OF 1	11738348 or 11738369
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Zinc chloride content	4.5.1	*	3.2	4.5.1
102	Ash content	4.5.2	*	3.3	4.5.2
103	Moisture content	4.5.3	*	3.5	4.5.3
104	Density	4.5.4	*	3.6	4.5.4
105	Hardness	4.5.5	*	3.7	4.5.5
106	Compressibility	4.5.6	*	3.8	4.5.6
107	Shear strength		(See 4.5.7)	3.9	4.5.7
108	Machining ability		(See 4.5.8)	3.10	4.5.8
109	Evidence of poor workmanship	4.4.3.1	*	3.12	Visual
110	Color	4.4.3.1	*	3.4	Visual

NOTES:

*Rejct the tubing lot on one defect.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing.-Testing is described in the Quality Conformance table.

4.4.3.1 Sample size.-From each lot of vulcanized fiber tubing (see 4.4.1) a minimum of 6 tubes shall be selected randomly just prior to the time of packaging or from the packaged tubes and subjected to the tests prescribed. When a lot of tubing consists of tubing from more than one leached batch, the sample shall contain a minimum of one sample from each leached batch. But in no case shall the sample size be less than six tubes to represent each lot. From each of the sample tubes selected, one 8 in. (203.2mm) long test sample shall be removed and sectionalized as indicated in figure 1. Specimens are to be taken from the cut end of the eight inch sample.

4.4.4 Inspection equipment.-The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the table para. 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection.-See 6.2 & 6.2.1.

4.5.1 Zinc chloride determination.-From each of the 8 in. (203.2mm) tube section submitted (see 4.4.3.1) a test specimen shall be taken and consist of five grams of finely divided particles, such as millings or filings of the material. The test specimen shall be dried for two hours at 105°C to 110°C. The dried specimen shall be placed in a Soxhlet, Wiley - Richardson or similar type of extractor, and extracted with distilled water for three hours. The rate of extraction with the Wiley-Richardson size of extractor shall be sufficient to cause the tube containing the sample to be filled by condensation and emptied by the siphone, at least six times per hour. When the Soxhlet or similar size extractor is used, the tube shall be filled and emptied at the rate of at least 3 cycles per hour. After this extract has cooled, 0.5 milliliter of potassium chromate solution (10 percent) shall be added, and the extract titrated with standard silver nitrate solution. The end point will be indicated by the appearance of a permanent red color. A blank test shall be made on an equal volume of distilled water and potassium chromate solution, and the result shall be subtracted from that of the specimen. The net result shall be calculated in terms of zinc chloride.

4.5.2 Ash content.-From each of the 8 in. (203.2mm) tube section submitted (see 4.4.3.1) a test specimen shall be taken and consist of two to five grams of finely divided particles, such as millings or filings of the material. The test specimen shall be dried for 20 hours at 100°C to 105°C cooled in the desiccator at room

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temperature, and weighed. The sample shall then be completely ashed in a muffle furnace at 750°C and the residue weighed. The ash content shall be expressed as a percentage, by weight, and shall be taken as the ratio of the weight of the residue to the weight of the dried specimen.

4.5.3 Moisture content.-One test specimen taken from each tube selected in accordance to 4.4.3.1 (see detail 1 of figure 1) shall be weighed to the nearest one milligram, then placed in a mechanical convection type oven and heated at a temperature of 135°C ± 2°C for 4 hours. At the end of the 4 hour drying period the specimens shall be removed from the oven and placed in a desiccator over anhydrous calcium chloride until cool. After cooling each specimen shall again be weighed to the nearest one milligram. The difference between the original weight of the specimen and the final weight shall be considered as the moisture content of the specimen and shall be calculated as follows:

$$\text{Moisture Content, Percent} = \frac{W1 - W2}{W2} \times 100$$

Where W1 = Original weight of specimen
W2 = Final weight of specimen

4.5.4 Density.-One test specimen taken from each tube selected in accordance to 4.4.3.1 (see detail 5 of figure 1) shall be accurately weighed in air to the nearest .01 gram, then suspended in a beaker of water at room temperature of 23°C ± 2°C and weighed again to the nearest .01 gram. The density in grams per cubic centimeter shall be determined as follows:

$$\text{Density, } D = \frac{DL \times WA}{WA - WL}$$

Where D = Density of specimen, grams per cubic centimeter
DL = Density of liquid (when other than water), grams per cubic centimeter
WA = Weight of specimen in air, grams
WL = Weight of specimen in liquid, grams

When water is used "DL" becomes unity. Care shall be exercised to remove air bubbles surrounding the specimen when being weighed in water.

NOTE: After removal of test samples from the sample tubes, the lot of tubing should be packaged in moisture proof containers to eliminate the possibility of change of properties due to change in moisture content.

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4.5.5 Hardness.-One test specimen taken from each tube selected in accordance to 4.4.3.1 (see detail 3 of figure 1) shall be tested using a Rockwell Hardess Tester, utilizing the Rockwell "R" scale.

4.5.6 Compressibility.-Two test specimen from each tube selected in accordance to 4.4.3.1 (see detail 2 of figure 1) shall be taken from the midwall of the tube so that the sides shall be parallel to the axis of the tube. After removal, the specimen shall be carefully machined or ground so that the thickness of the specimen will not vary by more than .001 in. (.0254mm) the entire surface. Each specimen shall be placed flat between steel blocks or in a test fixture in such a manner that a deflectometer can be used to measure the compression. A load shall be applied normal to the laminations. The load shall be applied at a rate that will insure that the required load is reached within 30 to 60 seconds. When load deflection curves are developed, the initial departure of the curve from the abscissa shall be the point from which compressibility will be measured.

4.5.7 Shear strength.-One test specimen taken from each tube selected in accordance with 4.4.3.1 (see detail 4 of figure 1) shall be subjected to this test. The test specimen shall be placed in a test fixture similar to that shown in figure 2 and tested at a loading rate that will produce the failure of the specimen between 30 to 60 seconds after initial application of the load. The data reported for each specimen shall include actual dimensions, shearing load and shear strength. If two or more test specimens had a shear strength below 4000 psi (27.58 MPa), the lot shall be rejected. Retest will be allowed if one test specimen had a tensile strength between 4000 psi (27.6 MPa) to 3500 psi (24.13 MPa). In the retesting procedure another 3 tubes randomly selected from the leached batch that failed shall be used to provide 3 additional test specimens. If any retested specimen had a tensile strength below 4000 psi (27.58 MPa), the lot shall be rejected.

4.5.8 Machining ability.-One test specimen of convenient length, taken from each tube in accordance to 4.4.3.1 shall be subjected to turning, milling, drilling and threading. The turning, milling, drilling and threading operations shall simulate the current production process for machining vulcanized fiber rotating bands. If the material of any test specimen split, chip, burr or follow the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging

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5.1.1 Level A - Packaging shall be sufficient to afford protection against deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the suppliers commercial practice when such meets the requirements of this level. The maximum quantity units per pack shall be 10 tubes each. See Specification MIL-STD-1169.

5.2 Packing.-Packing shall be level A or B as specified (see 6.3).

5.2.1 Level A - Fiber tubes, packaged as specified (see 6.3) shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Classification
PPP-B-585	Class 2 or 3
PPP-B-587	Any Type or Class, GR A
MIL-B-46506	Any Type or Class, GR A
PPP-B-601	Overseas Type
PPP-B-621	Class 2

Shipping containers shall be provided with caseliners conforming to Specification MIL-L-10547. Caseliners shall be closed and sealed in accordance with the Appendix to Specification MIL-L-10547. Boxes shall be closed, strapped, or banded in accordance with the applicable box specification or appendix thereto. A chipboard or fiberboard liner 1/16 or more thick shall completely line the bottom, top, and sides (between case liner and box) of box to protect case liner from nails/staples, etc.

5.2.2 Level B - Fiber tubes, packaged as specified (see 6.3) shall be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Classification
PPP-B-585	Class 1
PPP-B-587	Any Type or Class, GR B
MIL-B-46506	Any Type or Class, GR B
PPP-B-591	Class 1, Domestic
PPP-B-601	Domestic Type
PPP-B-621	Class 1
PPP-B-636	Domestic Class
PPP-B-640	Class 1

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Shipping containers shall be provided with caseliners conforming to Specification MIL-L-10547, caseliners shall be closed and sealed in accordance with the appendix to Specification MIL-L-10547. Boxes shall be closed, strapped or banded in accordance with the applicable box specification or appendix thereto. When wirebound or wood boxes are used, a chipboard (or fiberboard) liner, 1/16 or more thick shall completely line the bottom, top and sides (between case liner and box) of box to protect case liner from the box nails/staples, etc. Additionally, the fiber tubes (packaged as specified and using a caseliner) may be packed in other shipping containers which will insure acceptance by common carrier and will afford protection against physical damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall conform to the uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.3 Marking.-In addition to any special marking required by the contract or order, interior packages and exteriod shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.-The vulcanized fiber tubing is intended for use as rotating bands on 105MM Kinetic energy tank ammunition.

6.2 Test room conditions.-All tests shall be performed in a room conditioned to a temperature of 23°C ± 2°C and a relative humidity of 50% ± 5%.

6.2.1 Test specimens conditioning-prior to actual testing

a. Test specimens used for the compressibility, hardness, and shear strength tests shall be preconditioned by heating in an oven at 50°C ± 3°C for 48 hours and allowed to cool in a desiccator over calcium chloride for a period of at least 3 hours. The cooling process shall be performed in a room conditioned to a temperature of 23°C ± 2°C and a relative humidity of 50% ± 4%.

b. Test specimens used in all remaining tests shall be preconditioned for at least 3 hours in a room conditioned to a temperature of 23°C ± 2°C and a relative humidity of 50% ± 4%.

6.3 Ordering data.-See MIL-A-48078.

6.4 Submission of inspection equipment designs for approval.-See MIL-A-48078. Submit equipment designs, as required to:

Commander
US Army Research and Development Command
ATTN: DRDAR-QAR-I
Dover, N.J. 07801

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6.5 Process control.-Parameters such as time applicable to each phase of processing, temperatures, concentrations, humidities, etc. shall be established and maintained to assure homogeneity of product during production.

6.6 Discoloration.-Discoloration is not allowed. However, visual standards may be established to permit some types of discoloration. In all cases the contractor shall be responsible for submitting a product that meets the established physical and mechanical properties. Any visual standard used must be agreeable between the government and the contractor.

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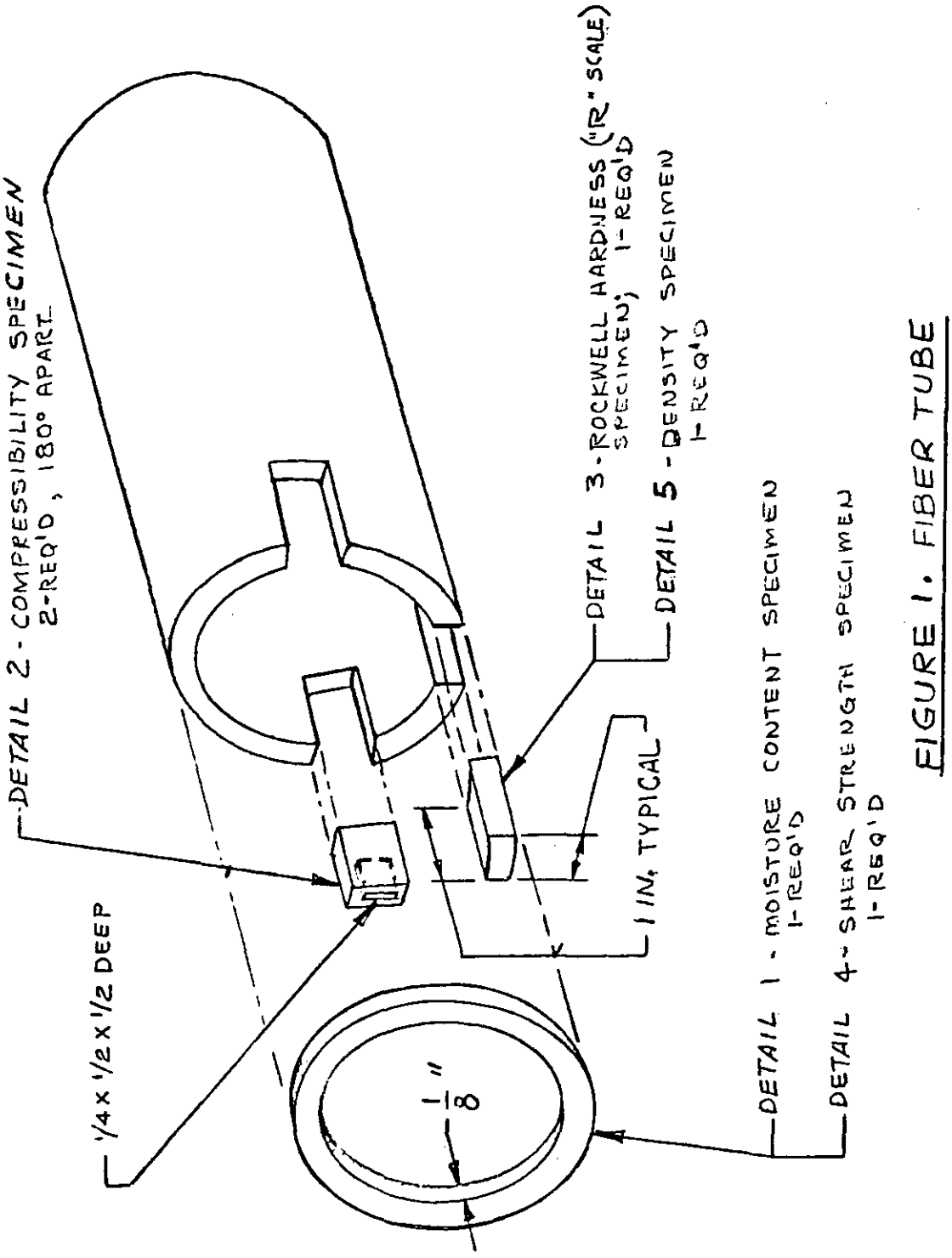
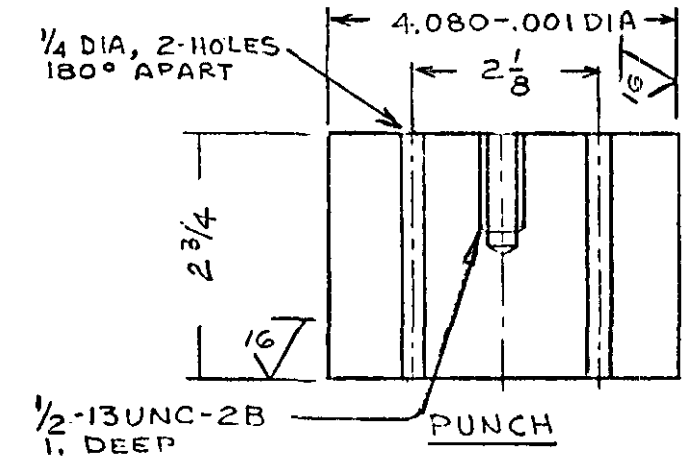
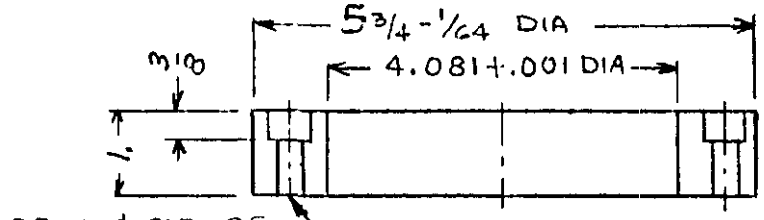
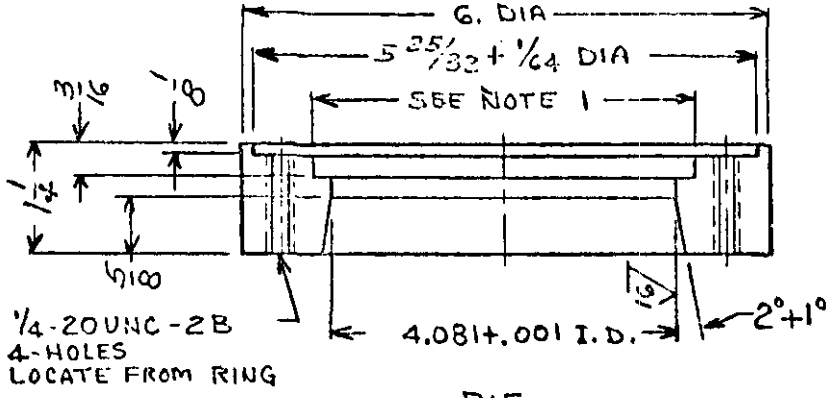


FIGURE 1. FIBER TUBE

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NOTES:
1- ADJUST DIA TO SUIT MFR'S SPECIMEN.
2- 125 √ FINISH, EXCEPT AS NOTED.
3- MAT'L :- STEEL, 4340, SPEC ASTM-A322.
4- HEAT TREATMENT - R_C 45-48.
5- REMOVE ALL BURRS.

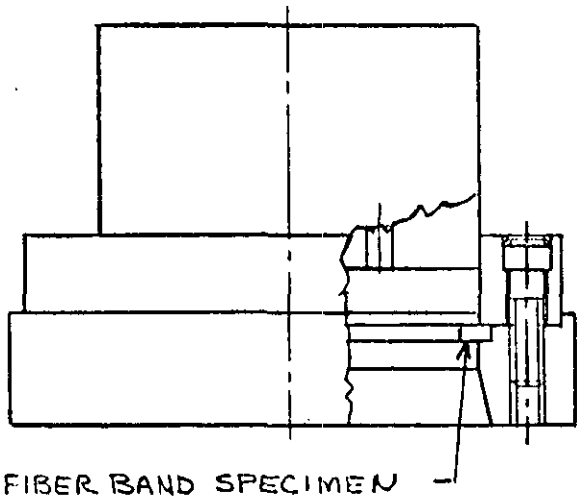


FIGURE 2 - SHEAR TEST FIXTURE

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-R-50792A Rotating Band Tubing, Vulcanized Fiber

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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